

Expt. No. : 14

Date :

3D PLOTTING

AIM

To write a python program to implement 3D plotting using matplotlib.

SOFTWARE REQUIRED

Python 3.14

THEORY

Introduction to 3D Plotting

A 3D plot allows you to represent data points or surfaces in three-dimensional space, with three axes: X, Y, and Z. This type of plot is particularly useful for visualizing multi-variable datasets and showing how data points are distributed or how a surface behaves in 3D space.

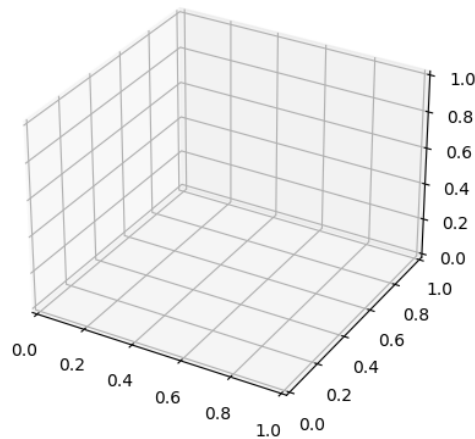
3D Plot Types

Common types of 3D plots in Matplotlib include:

- **3D Scatter Plot:** Displays data points in 3D space to show relationships between three variables.
- **3D Line Plot:** Plots a line in 3D space, used for showing trajectories or trends.
- **3D Surface Plot:** Creates a smooth surface in 3D space, useful for visualizing mathematical functions or surfaces.

Key Components of a 3D Plot

- **Axes:**
 - X-axis: Represents the independent variable along the horizontal axis.
 - Y-axis: Represents another independent variable along a perpendicular axis to the X-axis.
 - Z-axis: Represents the dependent variable, typically plotted vertically.
- **Figure and Axes Setup:** To create a 3D plot, you need to specify a 3D projection for the axes object, which is done using:
`fig = plt.figure()`
`ax = fig.add_subplot(111, projection='3d')`



Creating a 3D Plot in Matplotlib

To plot in 3D, the Axes3D object is used. Here's a general process:

1. **Importing required modules:** You need to import matplotlib.pyplot and mpl_toolkits.mplot3d for 3D plotting.
2. **Defining the data:** You define three arrays representing the X, Y, and Z coordinates.
3. **Creating the 3D plot:** Use functions like `ax.scatter()`, `ax.plot()`, or `ax.plot_surface()` to generate a 3D scatter, line, or surface plot, respectively.
 - **Customization:** You can add labels to the axes, change the viewpoint, and adjust colors, transparency, and grid lines.

PROBLEM

1. Build a Python program to create a 3D surface plot showing how speed and time affect distance travelled.
2. Build a Python program to create a 3D surface plot showing the variation of battery state of charge (SOC) with battery voltage and discharge current.
3. Madison is a little girl who is fond of toys. Her friend Mason works in a toy manufacturing factory. Mason has a 2D board A of size $H \times W$ with H rows and W columns. The board is divided into cells of size 1×1 with each cell indicated by its coordinate (i, j) . The cell (i, j) has an integer A_{ij} written on it. To create the toy Mason stacks A_{ij} number of cubes of size $1 \times 1 \times 1$ on the cell (i, j) . Given the description of the board showing the values of A_{ij} and that the price of the toy is equal to the 3d surface area find the price of the toy.

ALGORITHM

STEP 1: Start the program.

STEP 2: Open the file in the import for the matplotlib3d, numpy and matplotlib.pyplot.

STEP 3: Assign the plt.figure() in the value of fig.

STEP 4: Call the 3d function to perform the 3d projection.

STEP 5: Get input for three axis(x,y,z) to perform the operation.

STEP 6: Assign the operation for the three input axis(x,y,z).

STEP 7: Plot the value as per the given 3d axis value. As per you required color and access

the title. Finally show the output.

STEP 8: Stop the program.

PROGRAM CODING

```

import numpy as np
import matplotlib.pyplot as plt
n = int(input("Enter number of speed values: "))
speed = []
time = []
for i in range(n):
    s = float(input("Enter speed: "))
    t = float(input("Enter time: "))
    speed.append(s)
    time.append(t)
X, Y = np.meshgrid(speed, time)
Z = X * Y
fig = plt.figure()
ax = fig.add_subplot(111, projection="3d")
ax.plot_surface(X, Y, Z)
ax.set_xlabel("Speed")
ax.set_ylabel("Time")
ax.set_zlabel("Distance")
ax.set_title("Speed and Time vs Distance")
plt.show()

```

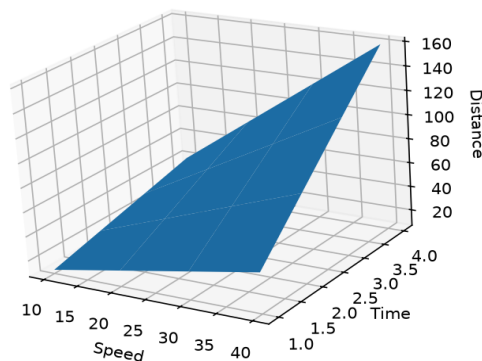
OUTCOMES OF PROGRAM CODING

```

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 6
4 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment
14.py =====
Enter number of speed values: 4
Enter speed: 10
Enter time: 1
Enter speed: 20
Enter time: 2
Enter speed: 30
Enter time: 3
Enter speed: 40
Enter time: 4

```

Speed and Time vs Distance



x=16.4341, y pane=4.2188, z=66.5034

PROBLEM CODING

```

import matplotlib.pyplot as plt
import numpy as np
n = int(input("Enter number of values: "))
voltage = []
current = []
for i in range(n):
    voltage.append(float(input("Enter battery voltage: ")))
    current.append(float(input("Enter discharge current: ")))
X, Y = np.meshgrid(voltage, current)
Z = 100 - (14 - X) * 20 - Y * 2
fig = plt.figure()
ax = fig.add_subplot(111, projection="3d")
ax.plot_surface(X, Y, Z)
ax.set_xlabel("Battery Voltage")
ax.set_ylabel("Discharge Current")
ax.set_zlabel("SOC")
plt.show()

```

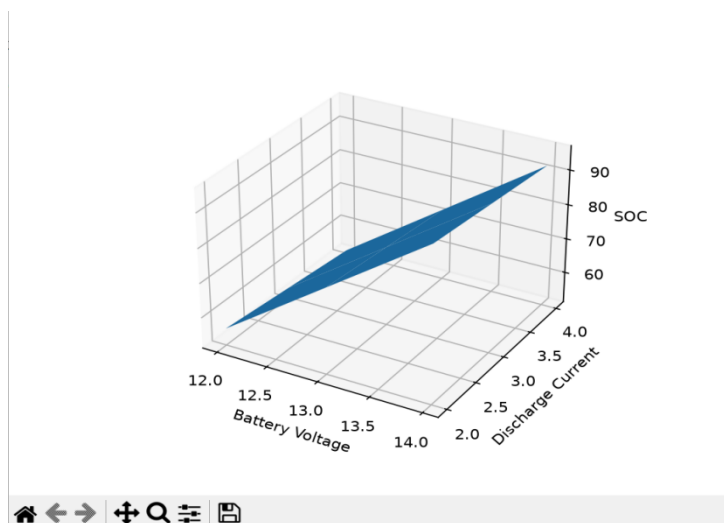
OUTCOMES OF PROGRAM CODING

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
 Enter "help" below or click "Help" above for more information.

>>>

===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment14.py =====

Enter number of values: 3
 Enter battery voltage: 12
 Enter discharge current: 2
 Enter battery voltage: 13
 Enter discharge current: 3
 Enter battery voltage: 14
 Enter discharge current: 4



PROBLEM CODING

```

h = int(input("Enter height of board: "))
w = int(input("Enter width of board: "))
a = []
for i in range(h):
    row = list(map(int, input("Enter row: ").split()))
    a.append(row)
area = 0
for i in range(h):
    for j in range(w):
        if a[i][j] > 0:
            area += 2
        if i == 0:
            area += a[i][j]
        else:
            area += max(0, a[i][j] - a[i-1][j])
        if i == h - 1:
            area += a[i][j]
        if j == 0:
            area += a[i][j]
        else:
            area += max(0, a[i][j] - a[i][j-1])
        if j == w - 1:
            area += a[i][j]
print("Surface Area =", area)

```

OUTCOMES OF PROGRAM CODING

```

IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem1\14.py =====
Enter height of board: 2
Enter width of board: 2
Enter row: 1 2
Enter row: 3 4
Surface Area = 34

```

VIVA QUESTION

1.What is the purpose of the mplot3d toolkit in Matplotlib?

The mplot3d toolkit in Matplotlib provides functions and utilities for creating 3D plots. It allows users to visualize data in three dimensions, enabling the representation of complex relationships among variables that are not easily captured in 2D plots.

2.Explain the significance of the command `ax = plt.axes(projection='3d')`.

The command `ax = plt.axes(projection='3d')` creates a 3D axis object that is used for plotting in three dimensions. This command sets up the coordinate system needed for 3D visualization, allowing users to plot data points, lines, or surfaces in a three-dimensional space.

3.Describe how to define the coordinates for a 3D plot using parametric equations.

In a 3D plot, coordinates are often defined using parametric equations, which express the X, Y, and Z coordinates as functions of a parameter, typically denoted as `t` or `z`. For example, to create a spiral, you might define:

- $x = z \cdot \sin(25z)$
- $y = z \cdot \cos(25z)$
- `z` can be defined as a linear range, such as `np.linspace(0, 2, 100)`. This way, as the parameter varies, the coordinates trace out a 3D path.

4.How can you customize the appearance of a 3D plot, such as changing colors and adding labels?

The appearance of a 3D plot can be customized using various Matplotlib functions:

- To change the color of a plot, specify the color in the plotting function, e.g., `ax.plot3D(x, y, z, color='blue')`.
- You can add axis labels using `ax.set_xlabel('X-axis')`, `ax.set_ylabel('Y-axis')`, and `ax.set_zlabel('Z-axis')`.
- A title can be added with `ax.set_title('Your Title Here')` to provide context for the plot.

5. What are some common types of 3D plots that can be created using Matplotlib?

Common types of 3D plots that can be created using Matplotlib include:

- **3D Scatter Plot:** Displays individual points in 3D space.
- **3D Line Plot:** Connects points with lines to show trends or paths in 3D.
- **3D Surface Plot:** Represents a surface in 3D space, useful for visualizing functions of two variables.
- **3D Wireframe Plot:** Displays a 3D surface as a mesh, highlighting the structure of the surface.
- **3D Contour Plot:** Shows contour lines on a 3D surface, useful for visualizing elevations or potential fields.

RESULT:

Thus, the Python program successfully generated a 3D plot representing the relationship between the parameters. The parameters were plotted on three axes, and the 3D visualization was displayed and verified successfully.

INFERENCE:

The experiment demonstrates that 3D plotting provides an effective way to visualize the relationship among three parameters simultaneously. It helps in identifying patterns, variations, and dependencies in data, thereby supporting better analysis and interpretation of their characteristics.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		

